

the time consuming process of design considerations, and limited scope for scalability and modularity in code when using a relational database.

- The distributed architecture of NoSQL databases ensures that they perform better than RDBMS. Besides, NoSQL doesn't have complex join operations and normalised data nor does it include complex queries. These factors give it the upper hand when it comes to database performance. Effective performance directly results in the better user experience of mobile apps because of the reduced load time of UI components and activities. This directly improves the battery life too.
- Security is another aspect that should never be ignored while trying to achieve all these goals. The databases should be able to communicate to the server over secure channels. Besides, the channels that communicate with mobile devices over the Internet demand low latency for an improved mobile user experience. Also, exchange of data on the network should be lightweight to meet these performance requirements.

All that said, we still may not be able to eliminate SQL databases which complement NoSQL in many ways. NoSQL doesn't guarantee atomicity or integrity of data which an RDBMS is capable of. So, it is the developer's needs at the end of day that decide which database to go with.

Integrating Couchbase Lite with Android Studio

Couchbase Lite is an open source project available under Apache License 2.0. It is an embedded JSON database that can work as a standalone, in a P2P network, or as a remote endpoint for a Sync Gateway. In this article, we explain how to power your Android apps with Couchbase Lite.

Before getting onto integration, let us check out a few key features about this database.

- Couchbase Lite stores and manages all the data locally on your mobile device in a lightweight JSON format.
- Considering the requirements for a mobile database, Couchbase Lite qualifies with its lower memory footprint, built-in security with user authentication, AES based data based encryption, and transport to the server through TLS.
- Couchbase Lite provides CRUD and query support through native APIs, and also works well with existing REST architectures with its programmatic access through REST API.
- Stream and batch APIs from Couchbase Lite enable the transfer of real-time data in batches with low network latency and throughput, thereby addressing the exact demands of mobile apps.

Let us now go through the steps of installing Couchbase Lite and the other basic operations to get started.

It is assumed that readers are already conversant with the Android Studio IDE for developing Android apps. Integrating Couchbase dB with Android is straightforward. You could start off by adding the dependency elements given below in your application's *build.gradle*:

```
dependencies {
    compile 'com.couchbase.lite:couchbase-lite-android:+'
}
```

In the Java part of the application, you would need the following basic set of packages to start with:

```
import com.couchbase.lite.*;
import com.couchbase.lite.android.AndroidContext;
```

Now that you are all set to use CouchBase APIs in your Android app, I'd like to illustrate a sample code, used for creating a database, and do an insert, update and delete of a document in it as mentioned in (1), (2), (3) and (4), respectively.

```
// (1) Get the database or create it if it doesn't already exist.
Manager manager = new Manager(new JavaContext(), Manager.DEFAULT_OPTIONS);
Database db = manager.getDatabase("couchdb");

// (2) Create a new document (a record) in the database.
Document doc = db.createDocument();
Map properties = new HashMap();
properties.put("firstName", "OSFY");
doc.putProperties(properties);

// (3) Update a document.
doc.update(new Document.DocumentUpdater() {
    @Override
    public boolean update(UnsavedRevision newRevision) {
        Map properties = newRevision.getUserProperties();
        properties.put("firstName", "Johnny");
        newRevision.setUserProperties(properties);
        return true;
    }
});

// (4) Delete a document.
doc.delete();
```



References

- [1] <https://developer.couchbase.com/mobile/>
- [2] <http://www.thegeekstuff.com/2014/01/sql-vs-nosql-db>
- [3] <https://android-arsenal.com/details/1/841>
- [4] <https://developer.couchbase.com/documentation/mobile/1.2/developer/guides/couchbase-lite/native-api/database/index.html>

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